Milestone 2

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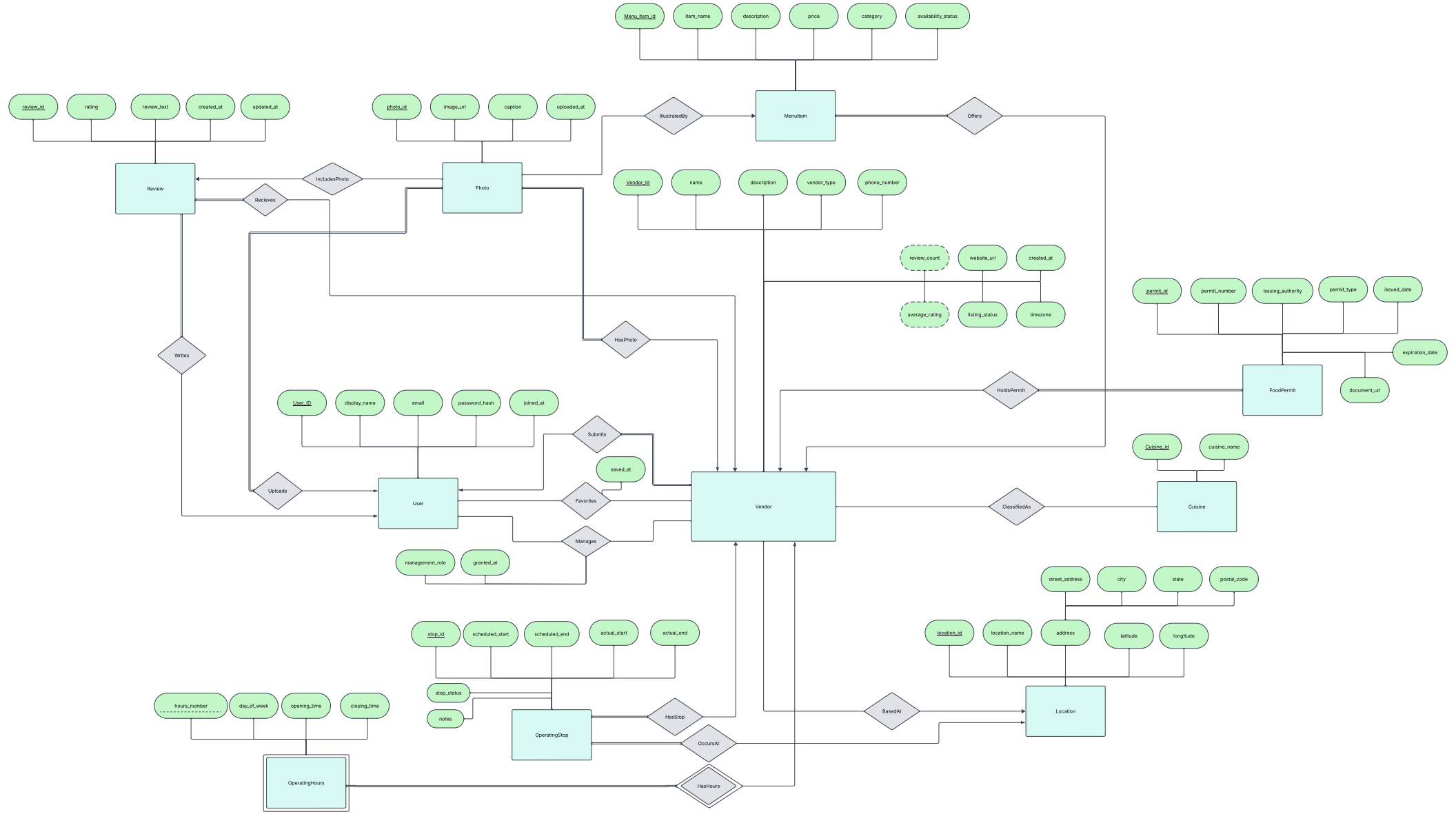
Part 1: Database Server

Our database, `restaurants`, is hosted at `35.245.147.165` and runs MySQL `8.4.11-google`. Connections use TLS with `--ssl-mode=REQUIRED`. All 16 tables use InnoDB.

Part 2: Table Schemas

Final E-R Diagram

We kept the Milestone 1 design and all 16 tables. The E-R diagram is unchanged.


Menu_item_id
item_name
description
price
category
availability_status
review_id
rating
review_text
created_at
updated_at
photo_id
image_url
caption
uploaded_at
IllustratedBy
MenuItem
Offers
Review
IncludesPhoto
Photo
Recieves
Vendor_id
name
description
vendor_type
phone_number
review_count
website_url
created_at
average_rating
listing_status
timezone
permit_id
permit_number
issuing_authority
permit_type
issued_date
expiration_date
document_url
HasPhoto
Writes
HoldsPermit
FoodPermit
User_ID
display_name
email
password_hash
joined_at
Submits
saved_at
Uploads
User
Favorites
Vendor
ClassifiedAs
Cuisine_id
cuisine_name
Cuisine
Manages
management_role
granted_at
street_address
city
state
postal_code
location_id
location_name
address
latitude
longitude
stop_id
scheduled_start
scheduled_end
actual_start
actual_end
stop_status
notes
OperatingStop
HasStop
BasedAt
Location
hours_number
day_of_week
opening_time
closing_time
OccursAt
OperatingHours
HasHours

Final Normalized Schemas (Unchanged)

User(user_id, display_name, email, password_hash, joined_at)

Vendor(vendor_id, name, description, vendor_type, phone_number, website_url, listing_status, created_at, timezone, submitted_by_user_id)

Cuisine(cuisine_id, cuisine_name)

Location(location_id, location_name, street_address, city, state, postal_code, latitude, longitude)

OperatingStop(stop_id, vendor_id, location_id, scheduled_start, scheduled_end, actual_start, actual_end, stop_status, notes)

OperatingHours(vendor_id, hours_number, day_of_week, opening_time, closing_time)

MenuItem(menu_item_id, vendor_id, item_name, description, price, category, availability_status)

Review(review_id, author_user_id, vendor_id, rating, review_text, created_at, updated_at)

Photo(photo_id, uploader_user_id, vendor_id, image_url, caption, uploaded_at)

FoodPermit(permit_id, vendor_id, permit_number, issuing_authority, permit_type, issued_date, expiration_date, document_url)

Manages(user_id, vendor_id, management_role, granted_at)

Favorites(user_id, vendor_id, saved_at)

VendorCuisine(vendor_id, cuisine_id)

VendorBase(vendor_id, location_id)

ReviewPhoto(photo_id, review_id)

MenuItemPhoto(photo_id, menu_item_id)

Implementation Decisions

- Primary and foreign keys follow the Milestone 1 relationships.
- OperatingHours uses weekday names and allows multiple shifts per day through its `(vendor_id, hours_number)` key.
- Average ratings and review counts are calculated from reviews.
- Imported authors have anonymous profiles, placeholder emails, and disabled password markers. Demo accounts use bcrypt password hashes.

- Imported listings belong to the Dataset Importer profile. Management access is recorded separately in Manages.
- Timestamps use UTC, while weekly opening hours use local time. Coordinates are rounded to six decimal places.

Table Schema and Row-Count Screenshots

These screenshots show `DESC` and the row count for each of the 16 tables.

User

```
mysql> DESC User;
+---------------+--------------+------+-----+-------------------+-------------------+
| Field         | Type         | Null | Key | Default           | Extra             |
+---------------+--------------+------+-----+-------------------+-------------------+
| user_id       | int          | NO   | PRI | NULL              | auto_increment    |
| display_name  | varchar(100) | NO   |     | NULL              |                   |
| email         | varchar(255) | NO   | UNI | NULL              |                   |
| password_hash | varchar(255) | NO   |     | NULL              |                   |
| joined_at     | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.275 sec)

mysql> SELECT COUNT(*) AS row_count FROM User;
+-----------+
| row_count |
+-----------+
|       146 |
+-----------+
1 row in set (0.478 sec)
```

Figure 1: User table structure and row count

Vendor

```
mysql> DESC Vendor;
+---------------------+--------------+------+-----+-------------------+-------------------+
| Field               | Type         | Null | Key | Default           | Extra             |
+---------------------+--------------+------+-----+-------------------+-------------------+
| vendor_id           | int          | NO   | PRI | NULL              | auto_increment    |
| name                | varchar(150) | NO   |     | NULL              |                   |
| description         | text         | YES  |     | NULL              |                   |
| vendor_type         | varchar(50)  | NO   |     | NULL              |                   |
| phone_number        | varchar(30)  | YES  |     | NULL              |                   |
| website_url         | varchar(500) | YES  |     | NULL              |                   |
| listing_status      | varchar(30)  | NO   |     | NULL              |                   |
| created_at          | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| timezone            | varchar(50)  | NO   |     | NULL              |                   |
| submitted_by_user_id | int          | NO   | MUL | NULL              |                   |
+---------------------+--------------+------+-----+-------------------+-------------------+
10 rows in set (0.593 sec)

mysql> SELECT COUNT(*) AS row_count FROM Vendor;
+-----------+
| row_count |
+-----------+
|        28 |
+-----------+
1 row in set (0.449 sec)
```

Figure 2: Vendor table structure and row count

Cuisine

```
mysql> DESC Cuisine;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| cuisine_id   | int          | NO   | PRI | NULL    | auto_increment |
| cuisine_name | varchar(100) | NO   | UNI | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
2 rows in set (0.356 sec)

mysql> SELECT COUNT(*) AS row_count FROM Cuisine;
+-----------+
| row_count |
+-----------+
|        14 |
+-----------+
1 row in set (0.479 sec)
```

Figure 3: Cuisine table structure and row count

Location

```
mysql> DESC Location;
+----------------+-------------+------+-----+---------+----------------+
| Field          | Type        | Null | Key | Default | Extra          |
+----------------+-------------+------+-----+---------+----------------+
| location_id    | int         | NO   | PRI | NULL    | auto_increment |
| location_name  | varchar(150) | YES  |     | NULL    |                |
| street_address | varchar(255) | YES  |     | NULL    |                |
| city           | varchar(100) | YES  |     | NULL    |                |
| state          | varchar(50)  | YES  |     | NULL    |                |
| postal_code    | varchar(20)  | YES  |     | NULL    |                |
| latitude       | decimal(9,6) | NO   |     | NULL    |                |
| longitude      | decimal(9,6) | NO   |     | NULL    |                |
+----------------+-------------+------+-----+---------+----------------+
8 rows in set (0.560 sec)

mysql> SELECT COUNT(*) AS row_count FROM Location;
+-----------+
| row_count |
+-----------+
|        25 |
+-----------+
1 row in set (0.560 sec)
```

Figure 4: Location table structure and row count

OperatingHours

```
mysql> DESC OperatingHours;
+--------------+-----------------------------------------------------------------------------------+------+-----+---------+--
-----+
| Field        | Type                                                                              | Null | Key | Default | E
xtra |
+--------------+-----------------------------------------------------------------------------------+------+-----+---------+--
-----+
| vendor_id    | int                                                                               | NO   | PRI | NULL    |
     |
| hours_number | int                                                                               | NO   | PRI | NULL    |
     |
| day_of_week  | enum('Monday','Tuesday','Wednesday','Thursday','Friday','Saturday','Sunday') | NO   |     | NULL    |
     |
| opening_time | time                                                                              | NO   |     | NULL    |
     |
| closing_time | time                                                                              | NO   |     | NULL    |
     |
+--------------+-----------------------------------------------------------------------------------+------+-----+---------+--
-----+
5 rows in set (0.419 sec)

mysql> SELECT COUNT(*) AS row_count FROM OperatingHours;
+-----------+
| row_count |
+-----------+
|       185 |
+-----------+
1 row in set (0.259 sec)
```

Figure 5: OperatingHours table structure and row count

OperatingStop

```
mysql> DESC OperatingStop;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| stop_id         | int         | NO   | PRI | NULL    | auto_increment |
| vendor_id       | int         | NO   | MUL | NULL    |                |
| location_id     | int         | NO   | MUL | NULL    |                |
| scheduled_start | datetime    | NO   |     | NULL    |                |
| scheduled_end   | datetime    | NO   |     | NULL    |                |
| actual_start    | datetime    | YES  |     | NULL    |                |
| actual_end      | datetime    | YES  |     | NULL    |                |
| stop_status     | varchar(30) | NO   |     | NULL    |                |
| notes           | text        | YES  |     | NULL    |                |
+-----------------+-------------+------+-----+---------+----------------+
9 rows in set (0.354 sec)

mysql> SELECT COUNT(*) AS row_count FROM OperatingStop;
+-----------+
| row_count |
+-----------+
|       112 |
+-----------+
1 row in set (0.402 sec)
```

Figure 6: OperatingStop table structure and row count

MenuItem

```
mysql> DESC MenuItem;
+--------------------+---------------+------+-----+---------+----------------+
| Field              | Type          | Null | Key | Default | Extra          |
+--------------------+---------------+------+-----+---------+----------------+
| menu_item_id       | int           | NO   | PRI | NULL    | auto_increment |
| vendor_id          | int           | NO   | MUL | NULL    |                |
| item_name          | varchar(150)  | NO   |     | NULL    |                |
| description        | text          | YES  |     | NULL    |                |
| price              | decimal(10,2) | NO   |     | NULL    |                |
| category           | varchar(100)  | YES  |     | NULL    |                |
| availability_status | varchar(30)   | NO   |     | NULL    |                |
+--------------------+---------------+------+-----+---------+----------------+
7 rows in set (0.212 sec)

mysql> SELECT COUNT(*) AS row_count FROM MenuItem;
+-----------+
| row_count |
+-----------+
|        84 |
+-----------+
1 row in set (0.257 sec)
```

Figure 7: MenuItem table structure and row count

Review

```
mysql> DESC Review;
+----------------+-----------+------+-----+-------------------+-------------------+
| Field          | Type      | Null | Key | Default           | Extra             |
+----------------+-----------+------+-----+-------------------+-------------------+
| review_id      | int       | NO   | PRI | NULL              | auto_increment    |
| author_user_id | int       | NO   | MUL | NULL              |                   |
| vendor_id      | int       | NO   | MUL | NULL              |                   |
| rating         | int       | NO   |     | NULL              |                   |
| review_text    | text      | NO   |     | NULL              |                   |
| created_at     | timestamp | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| updated_at     | timestamp | YES  |     | NULL              |                   |
+----------------+-----------+------+-----+-------------------+-------------------+
7 rows in set (0.370 sec)

mysql> SELECT COUNT(*) AS row_count FROM Review;
+-----------+
| row_count |
+-----------+
|       161 |
+-----------+
1 row in set (0.556 sec)
```

Figure 8: Review table structure and row count

Photo

```
mysql> DESC Photo;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| photo_id         | int          | NO   | PRI | NULL              | auto_increment    |
| uploader_user_id | int          | NO   | MUL | NULL              |                   |
| vendor_id        | int          | NO   | MUL | NULL              |                   |
| image_url        | varchar(500) | NO   |     | NULL              |                   |
| caption          | varchar(255) | YES  |     | NULL              |                   |
| uploaded_at      | timestamp    | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+--------------+------+-----+-------------------+-------------------+
6 rows in set (0.434 sec)

mysql> SELECT COUNT(*) AS row_count FROM Photo;
+-----------+
| row_count |
+-----------+
|       241 |
+-----------+
1 row in set (0.699 sec)
```

Figure 9: Photo table structure and row count

FoodPermit

```
mysql> DESC FoodPermit;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| permit_id         | int          | NO   | PRI | NULL    | auto_increment |
| vendor_id         | int          | NO   | MUL | NULL    |                |
| permit_number     | varchar(100) | NO   |     | NULL    |                |
| issuing_authority | varchar(150) | NO   | MUL | NULL    |                |
| permit_type       | varchar(100) | NO   |     | NULL    |                |
| issued_date       | date         | YES  |     | NULL    |                |
| expiration_date   | date         | YES  |     | NULL    |                |
| document_url      | varchar(500) | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
8 rows in set (1.140 sec)

mysql> SELECT COUNT(*) AS row_count FROM FoodPermit;
+-----------+
| row_count |
+-----------+
|        28 |
+-----------+
1 row in set (0.748 sec)
```

Figure 10: FoodPermit table structure and row count

Manages

```
mysql> DESC Manages;
+-----------------+-------------+------+-----+-------------------+-------------------+
| Field           | Type        | Null | Key | Default           | Extra             |
+-----------------+-------------+------+-----+-------------------+-------------------+
| user_id         | int         | NO   | PRI | NULL              |                   |
| vendor_id       | int         | NO   | PRI | NULL              |                   |
| management_role | varchar(50) | NO   |     | NULL              |                   |
| granted_at      | timestamp   | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----------------+-------------+------+-----+-------------------+-------------------+
4 rows in set (0.800 sec)

mysql> SELECT COUNT(*) AS row_count FROM Manages;
+-----------+
| row_count |
+-----------+
|        30 |
+-----------+
1 row in set (0.411 sec)
```

Figure 11: Manages table structure and row count

Favorites

```
mysql> DESC Favorites;
+-----------+-----------+------+-----+-------------------+-------------------+
| Field     | Type      | Null | Key | Default           | Extra             |
+-----------+-----------+------+-----+-------------------+-------------------+
| user_id   | int       | NO   | PRI | NULL              |                   |
| vendor_id | int       | NO   | PRI | NULL              |                   |
| saved_at  | timestamp | NO   |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----------+-----------+------+-----+-------------------+-------------------+
3 rows in set (0.422 sec)

mysql> SELECT COUNT(*) AS row_count FROM Favorites;
+-----------+
| row_count |
+-----------+
|        60 |
+-----------+
1 row in set (0.258 sec)
```

Figure 12: Favorites table structure and row count

VendorCuisine

```
mysql> DESC VendorCuisine;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| vendor_id  | int  | NO   | PRI | NULL    |       |
| cuisine_id | int  | NO   | PRI | NULL    |       |
+------------+------+------+-----+---------+-------+
2 rows in set (0.252 sec)

mysql> SELECT COUNT(*) AS row_count FROM VendorCuisine;
+-----------+
| row_count |
+-----------+
|        35 |
+-----------+
1 row in set (0.376 sec)
```

Figure 13: VendorCuisine table structure and row count

VendorBase

```
mysql> DESC VendorBase;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| vendor_id   | int  | NO   | PRI | NULL    |       |
| location_id | int  | NO   | MUL | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.383 sec)

mysql> SELECT COUNT(*) AS row_count FROM VendorBase;
+-----------+
| row_count |
+-----------+
|        25 |
+-----------+
1 row in set (0.233 sec)
```

Figure 14: VendorBase table structure and row count

ReviewPhoto

```
mysql> DESC ReviewPhoto;
+-----------+------+------+-----+---------+-------+
| Field     | Type | Null | Key | Default | Extra |
+-----------+------+------+-----+---------+-------+
| photo_id  | int  | NO   | PRI | NULL    |       |
| review_id | int  | NO   | MUL | NULL    |       |
+-----------+------+------+-----+---------+-------+
2 rows in set (0.316 sec)

mysql> SELECT COUNT(*) AS row_count FROM ReviewPhoto;
+-----------+
| row_count |
+-----------+
|       157 |
+-----------+
1 row in set (0.313 sec)
```

Figure 15: ReviewPhoto table structure and row count

MenuItemPhoto

```
mysql> DESC MenuItemPhoto;
+--------------+------+------+-----+---------+-------+
| Field        | Type | Null | Key | Default | Extra |
+--------------+------+------+-----+---------+-------+
| photo_id     | int  | NO   | PRI | NULL    |       |
| menu_item_id | int  | NO   | MUL | NULL    |       |
+--------------+------+------+-----+---------+-------+
2 rows in set (0.455 sec)

mysql> SELECT COUNT(*) AS row_count FROM MenuItemPhoto;
+-----------+
| row_count |
+-----------+
|        84 |
+-----------+
1 row in set (0.695 sec)
```

Figure 16: MenuItemPhoto table structure and row count

Part 3: Dataset and Population

Sources and Preparation

We used UCSD's **Google Local Data (2021)**, specifically the Virginia business metadata and 10-core reviews, instead of the proposal's suggested Yelp dataset. Python filtered food businesses, matched reviews by `gmap_id`, and split the records into our tables. Source IDs were mapped to database IDs to keep the relationships connected.

The dataset contains **25 real businesses and 3 fictional mobile/pop-up vendors**. The imported information is historical. We loaded the data using 16 multi-row INSERT statements, with referenced records added before dependent records.

Real and Synthetic Data

The dataset includes **105 real reviews and 56 labeled demo reviews**. Real review text and ratings are preserved. Imported authors are anonymized and have placeholder emails because they did not register for our app.

Business details, locations, cuisines, and weekly hours come from the source where available. Menu items, prices, permits, managers, favorites, and dated operating stops are synthetic. Missing schedules and contact details use demo placeholders. Fictional content is labeled in the records.

Generic menu photos come from TheMealDB and are illustrations. Permit numbers starting with `DEMO-NOT-VALID` are fictional. Unknown optional values, including permit documents and actual times for future stops, remain NULL.

Row Counts and Practical Coverage

These counts were checked against the database and stayed unchanged after the rolled-back tests.

Table	Rows
User	146
Vendor	28
Cuisine	14
Location	25
OperatingHours	185
OperatingStop	112
MenuItem	84
Review	161

Table	Rows
Photo	241
FoodPermit	28
Manages	30
Favorites	60
VendorCuisine	35
VendorBase	25
ReviewPhoto	157
MenuItemPhoto	84

Each vendor has three menu items and four dated stops. Twelve demo customers each have five favorites. Vendors share cuisine labels, reviews link to authors, and photos link to reviews and menu items. The three mobile vendors also give us examples without a permanent base location.

These records let us test searches, profiles, reviews, menus, schedules, and favorites across related tables. Part 2 includes the schema and count screenshots.

Representative Records

The samples below show records from each table. Long text is shortened, and password hashes are omitted.

User

user_id	display_name	joined_at
1	Dataset Importer	2026-10-23 03:43:11
25	Imported User 00024	2026-10-23 03:43:11

Vendor

vendor_id	name	vendor_type
1	Adela's Deli	restaurant
2	Maggios	restaurant

Cuisine

cuisine_id	cuisine_name
1	Italian
2	Brazilian

Location

location_id	city	latitude	longitude
1	Williamsburg	37.207971	-76.610804
2	Alexandria	38.794352	-77.070114

OperatingHours

vendor_id	day_of_week	opening_time	closing_time
1	Monday	11:00:00	20:00:00
1	Tuesday	11:00:00	20:00:00

OperatingStop

stop_id	vendor_id	location_id	stop_status
1	1	1	demo_completed
2	1	1	demo_completed

MenuItem

menu_item_id	vendor_id	item_name	price
1	1	[DEMO] kofta burgers	15.88
2	1	[DEMO] Pancakes	21.42

Review

review_id	vendor_id	rating	review_text
24	20	5	Title says it all. Big selection of combo lunches for less than $5.00…
25	20	1	Okay I totally love my regular place, but decided to check this one o…

Photo

photo_id	vendor_id	uploader_user_id	caption
4	20	26	[IMPORTED] Source review photo; no caption supplied.
268	8	1511	[IMPORTED] Source review photo; no caption supplied.

FoodPermit

permit_id	vendor_id	permit_number
1	1	DEMO-NOT-VALID-0001
2	2	DEMO-NOT-VALID-0002

Manages

user_id	vendor_id	management_role
13866	1	demo_owner
13867	2	demo_owner

Favorites

user_id	vendor_id	saved_at
13947	7	2026-10-23 03:43:11
13947	13	2026-10-23 03:43:11

VendorCuisine

vendor_id	cuisine_id
1	24
2	1

VendorBase

vendor_id	location_id
1	1
2	2

ReviewPhoto

photo_id	review_id
4	25
268	1597

MenuItemPhoto

photo_id	menu_item_id
1541	1
1542	2

Part 4: SQL Commands and Verification

Regular SQL Commands

The SQL below covers table creation and the queries our app needs. Full commands are included in the appendices.

The regular SQL covers the five required categories:

Category	Commands
Create tables	The 16 CREATE TABLE statements in Appendix A
Retrieve data	Queries 1-24
Add data	Registration, vendor, location, menu, review, photo, favorite and permit inserts
Update data	Profile, contact details, menu price, review, hours, stop and permit updates
Delete data	Favorite, review, menu-item and photo deletions

A vendor listing is taken down by setting its status to `inactive`, preserving its associated reviews and other records.

Application action	Query numbers
Browse, search, filter, and sort	1-7
Vendor details, cuisines, menus, photos, hours, stops, reviews, permits	8-16
Favorites, user profiles/reviews, managed vendors, login lookup	17-21
Owner review, profile, and operating-log exports	22-24
Register/edit account, submit vendor, assign approved manager	25-28
Add location/base, cuisine, and weekly hours	29-31
Update stops, hours, menus, business details, reviews and photos	32-36
Favorites, permits, review/menu deletion, listing removal, photo deletion	37-42
Menu/schedule imports	Reuse 33 or 32 for validated rows in one transaction

The queries use sample values. The write examples end with ROLLBACK so the practice changes are not saved.

Advanced SQL: CHECK Constraints

The implemented checks are:

Table	Rule
Review	Rating is between 1 and 5
MenuItem	Price is nonnegative
Location	Latitude is between -90 and 90
Location	Longitude is between -180 and 180
OperatingStop	Scheduled end is after scheduled start
OperatingStop	Actual end is NULL, or actual start exists and precedes it

We use CHECK constraints to reject values that break these rules. This is our first advanced SQL feature.

Advanced SQL: Triggers

Our second advanced SQL feature is triggers. A photo linked to a review or menu item must belong to the same vendor. The triggers reject links that break this rule and prevent later updates from making an existing link invalid.

There are seven triggers because the rule needs to be checked when adding or updating links and when changing the vendor on a photo, review, or menu item. The full definitions are in Appendix C, with tests and results in Appendix D.

Recorded Tests

All 42 final query groups ran without MySQL errors. Eighteen checks confirmed that the practice inserts, updates, and deletes produced the expected results. After ROLLBACK, row counts and stored data were unchanged across all 16 tables.

We also tested valid and invalid photo links and an out-of-range review rating. The results are listed below.

Test	Recorded outcome
Final app queries (1-42)	Completed without MySQL errors

Test	Recorded outcome
Valid same-vendor review-photo attachment	Accepted, then rolled back
Valid same-vendor menu-photo attachment	Accepted, then rolled back
Seven cross-vendor attachment/vendor changes	Rejected with SQLSTATE 45000
Practice write checks	All 18 passed with triggers installed
Data after rollback	Counts and stored contents unchanged in all 16 tables
Earlier rating-6 CHECK test	Rejected with error 3819

Appendix D includes the trigger tests and their recorded output. The schema/count screenshots are included in Part 2.

Query Plan Observation

We ran EXPLAIN on the query joining Review and User for vendor 1. MySQL used the `vendor_id` index for reviews and the primary key for users. The table shows its estimated row counts.

Table alias	Access type	Index	Estimated rows
r (Review)	ref	vendor_id	2
u (User)	eq_ref	PRIMARY	1

Sources

- UCSD McAuley Lab, *Google Local Data (2021)*. Virginia metadata and 10-core review files. https://mcauleylab.ucsd.edu/public_datasets/gdrive/googlelocal/
- TheMealDB, illustrative meal-photo metadata and image URLs. https://www.themealdb.com/docs_api_guide.php. Generic photographs are used only as labeled illustrations, not as photos of the real vendors' offerings.
- CS4750 Fall 2026, *Milestone 2: DB setup and SQL*, assignment instructions supplied with the project materials.

Appendix A: Table-Creation SQL

These statements define the 16 application tables.

```
CREATE DATABASE IF NOT EXISTS restaurants;
USE restaurants;

CREATE TABLE `User` (
  user_id INT AUTO_INCREMENT,
  display_name VARCHAR(100) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  joined_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (user_id)
);

CREATE TABLE Cuisine (
  cuisine_id INT AUTO_INCREMENT,
  cuisine_name VARCHAR(100) NOT NULL UNIQUE,
  PRIMARY KEY (cuisine_id)
);

CREATE TABLE Location (
  location_id INT AUTO_INCREMENT,
  location_name VARCHAR(150),
  street_address VARCHAR(255),
  city VARCHAR(100),
  state VARCHAR(50),
  postal_code VARCHAR(20),
  latitude DECIMAL(9,6) NOT NULL,
  longitude DECIMAL(9,6) NOT NULL,
  PRIMARY KEY (location_id),
  CONSTRAINT chk_latitude CHECK (latitude BETWEEN -90 AND 90),
  CONSTRAINT chk_longitude CHECK (longitude BETWEEN -180 AND 180)
);

CREATE TABLE Vendor (
  vendor_id INT AUTO_INCREMENT,
  name VARCHAR(150) NOT NULL,
  description TEXT,
  vendor_type VARCHAR(50) NOT NULL,
  phone_number VARCHAR(30),
  website_url VARCHAR(500),
```

```
  listing_status VARCHAR(30) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  timezone VARCHAR(50) NOT NULL,
  submitted_by_user_id INT NOT NULL,
  PRIMARY KEY (vendor_id),
  FOREIGN KEY (submitted_by_user_id) REFERENCES `User`(user_id)
);

CREATE TABLE OperatingStop (
  stop_id INT AUTO_INCREMENT,
  vendor_id INT NOT NULL,
  location_id INT NOT NULL,
  scheduled_start DATETIME NOT NULL,
  scheduled_end DATETIME NOT NULL,
  actual_start DATETIME,
  actual_end DATETIME,
  stop_status VARCHAR(30) NOT NULL,
  notes TEXT,
  PRIMARY KEY (stop_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  FOREIGN KEY (location_id) REFERENCES Location(location_id),
  CONSTRAINT chk_sched_order CHECK (scheduled_end > scheduled_start),
  CONSTRAINT chk_actual_order CHECK (actual_end IS NULL OR (actual_start IS
   ↪  NOT NULL AND actual_end > actual_start))
);

CREATE TABLE OperatingHours (
  vendor_id INT NOT NULL,
  hours_number INT NOT NULL,
  day_of_week ENUM('Monday','Tuesday','Wednesday','Thursday','Friday',
 ↪  'Saturday','Sunday') NOT NULL,
  opening_time TIME NOT NULL,
  closing_time TIME NOT NULL,
  PRIMARY KEY (vendor_id, hours_number),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id)
);

CREATE TABLE MenuItem (
  menu_item_id INT AUTO_INCREMENT,
  vendor_id INT NOT NULL,
  item_name VARCHAR(150) NOT NULL,
  description TEXT,
  price DECIMAL(10,2) NOT NULL,
```

```
  category VARCHAR(100),
  availability_status VARCHAR(30) NOT NULL,
  PRIMARY KEY (menu_item_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  CONSTRAINT chk_price CHECK (price >= 0)
);

CREATE TABLE Review (
  review_id INT AUTO_INCREMENT,
  author_user_id INT NOT NULL,
  vendor_id INT NOT NULL,
  rating INT NOT NULL,
  review_text TEXT NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NULL,
  PRIMARY KEY (review_id),
  FOREIGN KEY (author_user_id) REFERENCES `User`(user_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  CONSTRAINT chk_rating CHECK (rating BETWEEN 1 AND 5)
);

CREATE TABLE Photo (
  photo_id INT AUTO_INCREMENT,
  uploader_user_id INT NOT NULL,
  vendor_id INT NOT NULL,
  image_url VARCHAR(500) NOT NULL,
  caption VARCHAR(255),
  uploaded_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (photo_id),
  FOREIGN KEY (uploader_user_id) REFERENCES `User`(user_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id)
);

CREATE TABLE FoodPermit (
  permit_id INT AUTO_INCREMENT,
  vendor_id INT NOT NULL,
  permit_number VARCHAR(100) NOT NULL,
  issuing_authority VARCHAR(150) NOT NULL,
  permit_type VARCHAR(100) NOT NULL,
  issued_date DATE,
  expiration_date DATE,
  document_url VARCHAR(500),
  PRIMARY KEY (permit_id),
```

```
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  CONSTRAINT uq_permit UNIQUE (issuing_authority, permit_number)
);

CREATE TABLE Manages (
  user_id INT NOT NULL,
  vendor_id INT NOT NULL,
  management_role VARCHAR(50) NOT NULL,
  granted_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (user_id, vendor_id),
  FOREIGN KEY (user_id) REFERENCES `User`(user_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id)
);

CREATE TABLE Favorites (
  user_id INT NOT NULL,
  vendor_id INT NOT NULL,
  saved_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (user_id, vendor_id),
  FOREIGN KEY (user_id) REFERENCES `User`(user_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id)
);

CREATE TABLE VendorCuisine (
  vendor_id INT NOT NULL,
  cuisine_id INT NOT NULL,
  PRIMARY KEY (vendor_id, cuisine_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  FOREIGN KEY (cuisine_id) REFERENCES Cuisine(cuisine_id)
);

CREATE TABLE VendorBase (
  vendor_id INT NOT NULL,
  location_id INT NOT NULL,
  PRIMARY KEY (vendor_id),
  FOREIGN KEY (vendor_id) REFERENCES Vendor(vendor_id),
  FOREIGN KEY (location_id) REFERENCES Location(location_id)
);

CREATE TABLE ReviewPhoto (
  photo_id INT NOT NULL,
  review_id INT NOT NULL,
  PRIMARY KEY (photo_id),
```

```
  FOREIGN KEY (photo_id) REFERENCES Photo(photo_id),
  FOREIGN KEY (review_id) REFERENCES Review(review_id)
);

CREATE TABLE MenuItemPhoto (
  photo_id INT NOT NULL,
  menu_item_id INT NOT NULL,
  PRIMARY KEY (photo_id),
  FOREIGN KEY (photo_id) REFERENCES Photo(photo_id),
  FOREIGN KEY (menu_item_id) REFERENCES MenuItem(menu_item_id)
);
```

Appendix B: Application SQL

Queries 1-24 retrieve records. The write examples begin before Query 25 and end in ROLLBACK.

```
USE restaurants;

-- Query 1
SELECT vendor_id, name, vendor_type FROM Vendor ORDER BY name;

-- Query 2
SELECT vendor_id, name FROM Vendor
WHERE name LIKE '%pizza%' OR description LIKE '%pizza%';

-- Query 3
SELECT v.vendor_id, v.name
FROM Vendor v
JOIN VendorCuisine vc ON vc.vendor_id = v.vendor_id
JOIN Cuisine c ON c.cuisine_id = vc.cuisine_id
WHERE c.cuisine_name = 'Italian';

-- Query 4
SELECT vendor_id, name FROM Vendor
WHERE vendor_id IN (
    SELECT vb.vendor_id FROM VendorBase vb
    JOIN Location l ON l.location_id = vb.location_id
    WHERE l.city = 'Alexandria'
    UNION
    SELECT s.vendor_id FROM OperatingStop s
    JOIN Location l ON l.location_id = s.location_id
    WHERE l.city = 'Alexandria' AND s.scheduled_end >= UTC_TIMESTAMP()
      AND s.stop_status NOT IN ('cancelled', 'demo_cancelled')
);

-- Query 5
SELECT vendor_id, name FROM Vendor WHERE vendor_type = 'food_truck';

-- Query 6
SELECT v.vendor_id, v.name, ROUND(AVG(r.rating), 2) AS average_rating,
       COUNT(r.review_id) AS review_count
FROM Vendor v
LEFT JOIN Review r ON r.vendor_id = v.vendor_id
GROUP BY v.vendor_id, v.name
```

```sql
ORDER BY average_rating DESC;

-- Query 7
SELECT v.vendor_id, v.name, l.location_name,
       ST_Distance_Sphere(POINT(l.longitude, l.latitude),
                          POINT(-78.476678, 38.029306)) / 1000 AS distance_km
FROM Vendor v
JOIN (
    SELECT vendor_id, location_id FROM VendorBase
    UNION
    SELECT vendor_id, location_id FROM OperatingStop
    WHERE scheduled_start <= UTC_TIMESTAMP() AND scheduled_end >
     ↪  UTC_TIMESTAMP()
      AND stop_status NOT IN ('cancelled', 'demo_cancelled')
) locations ON locations.vendor_id = v.vendor_id
JOIN Location l ON l.location_id = locations.location_id
ORDER BY distance_km;

-- Query 8
SELECT v.*, l.street_address, l.city, l.state, l.postal_code
FROM Vendor v
LEFT JOIN VendorBase vb ON vb.vendor_id = v.vendor_id
LEFT JOIN Location l ON l.location_id = vb.location_id
WHERE v.vendor_id = 1;

-- Query 9
SELECT c.cuisine_name FROM Cuisine c
JOIN VendorCuisine vc ON vc.cuisine_id = c.cuisine_id
WHERE vc.vendor_id = 1;

-- Query 10
SELECT menu_item_id, item_name, description, price, category,
 ↪  availability_status
FROM MenuItem WHERE vendor_id = 1 ORDER BY price;

-- Query 11
SELECT m.item_name, p.image_url, p.caption
FROM MenuItem m
JOIN MenuItemPhoto mp ON mp.menu_item_id = m.menu_item_id
JOIN Photo p ON p.photo_id = mp.photo_id
WHERE m.vendor_id = 1;

-- Query 12
```

```sql
SELECT day_of_week, opening_time, closing_time
FROM OperatingHours WHERE vendor_id = 1
ORDER BY day_of_week, opening_time;

-- Query 13
SELECT s.*, l.location_name, l.city, l.latitude, l.longitude
FROM OperatingStop s
JOIN Location l ON l.location_id = s.location_id
WHERE s.vendor_id = 1 ORDER BY s.scheduled_start;

-- Query 14
SELECT r.review_id, u.display_name, r.rating, r.review_text, r.created_at
FROM Review r
JOIN User u ON u.user_id = r.author_user_id
WHERE r.vendor_id = 1 ORDER BY r.created_at DESC;

-- Query 15
SELECT rp.review_id, p.image_url, p.caption
FROM ReviewPhoto rp
JOIN Review r ON r.review_id = rp.review_id
JOIN Photo p ON p.photo_id = rp.photo_id
WHERE r.vendor_id = 1;

-- Query 16
SELECT * FROM FoodPermit WHERE vendor_id = 1;

-- Query 17
SELECT v.vendor_id, v.name, f.saved_at
FROM Favorites f JOIN Vendor v ON v.vendor_id = f.vendor_id
WHERE f.user_id = 13947;

-- Query 18
SELECT user_id, display_name, email, joined_at FROM User WHERE user_id =
 ↪  13947;

-- Query 19
SELECT r.review_id, v.name, r.rating, r.review_text
FROM Review r JOIN Vendor v ON v.vendor_id = r.vendor_id
WHERE r.author_user_id = 13947;

-- Query 20
SELECT v.vendor_id, v.name, m.management_role
FROM Manages m JOIN Vendor v ON v.vendor_id = m.vendor_id
```

```sql
WHERE m.user_id = 13866;

-- Query 21
SELECT user_id, display_name, password_hash FROM User
WHERE email = 'demo.customer.001@example.invalid'
  AND password_hash NOT LIKE '!IMPORTED_NO_LOGIN!%';

-- Query 22
SELECT r.review_id, r.rating, r.review_text, r.created_at
FROM Review r JOIN Manages m ON m.vendor_id = r.vendor_id
WHERE r.vendor_id = 1 AND m.user_id = 13866;

-- Query 23
SELECT v.* FROM Vendor v JOIN Manages m ON m.vendor_id = v.vendor_id
WHERE v.vendor_id = 1 AND m.user_id = 13866;

-- Query 24
SELECT s.* FROM OperatingStop s JOIN Manages m ON m.vendor_id = s.vendor_id
WHERE s.vendor_id = 1 AND m.user_id = 13866;

START TRANSACTION;

-- Query 25
INSERT INTO User (display_name, email, password_hash, joined_at)
VALUES ('Query Demo User', CONCAT('query-demo-', UUID(), '@example.invalid'),
        '$2y$10$OrsTj3NedJZ8oeMzYVdt2O8aZZW901CYshgDr75b2VBFGrtlBjzXS',
         ↪  UTC_TIMESTAMP());
SET @new_user_id = LAST_INSERT_ID();

-- Query 26
UPDATE User SET display_name = 'Updated Query Demo User' WHERE user_id =
 ↪  @new_user_id;

-- Query 27
INSERT INTO Vendor (name, description, vendor_type, phone_number,
 ↪  website_url,
                    listing_status, created_at, timezone,
 ↪  submitted_by_user_id)
VALUES ('[DEMO] Query Practice Vendor', 'Disposable SQL learning example',
 ↪  'food_truck',
        NULL, NULL, 'demo', UTC_TIMESTAMP(), 'America/New_York',
         ↪  @new_user_id);
```

```
SET @new_vendor_id = LAST_INSERT_ID();

-- Query 28
INSERT INTO Manages (user_id, vendor_id, management_role, granted_at)
VALUES (@new_user_id, @new_vendor_id, 'demo_owner', UTC_TIMESTAMP());

-- Query 29
INSERT INTO Location (location_name, street_address, city, state,
 ↪  postal_code, latitude, longitude)
VALUES ('[DEMO] Practice Stop', NULL, 'Charlottesville', 'VA', NULL,
 ↪  38.029306, -78.476678);
SET @new_location_id = LAST_INSERT_ID();
INSERT INTO VendorBase (vendor_id, location_id) VALUES (@new_vendor_id,
 ↪  @new_location_id);

-- Query 30
SET @existing_cuisine_id = (SELECT cuisine_id FROM Cuisine ORDER BY
 ↪  cuisine_id LIMIT 1);
INSERT INTO VendorCuisine (vendor_id, cuisine_id) VALUES (@new_vendor_id,
 ↪  @existing_cuisine_id);

-- Query 31
INSERT INTO OperatingHours (vendor_id, hours_number, day_of_week,
 ↪  opening_time, closing_time)
VALUES (@new_vendor_id, 1, 'Monday', '11:00:00', '19:00:00');

-- Query 32
INSERT INTO OperatingStop (vendor_id, location_id, scheduled_start,
 ↪  scheduled_end,
                           actual_start, actual_end, stop_status, notes)
SELECT @new_vendor_id, @new_location_id, UTC_TIMESTAMP() + INTERVAL 1 DAY,
       UTC_TIMESTAMP() + INTERVAL 1 DAY + INTERVAL 4 HOUR,
       NULL, NULL, 'demo_scheduled', '[DEMO] Query practice stop'
WHERE EXISTS (SELECT 1 FROM Manages WHERE vendor_id = @new_vendor_id AND
 ↪  user_id = @new_user_id);
SET @new_stop_id = LAST_INSERT_ID();
UPDATE OperatingStop s
SET s.notes = '[DEMO] Updated practice stop'
WHERE s.stop_id = @new_stop_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = s.vendor_id AND
   ↪  m.user_id = @new_user_id);
```

```
UPDATE OperatingStop s
SET s.location_id = @new_location_id,
    s.scheduled_start = UTC_TIMESTAMP() + INTERVAL 2 DAY,
    s.scheduled_end = UTC_TIMESTAMP() + INTERVAL 2 DAY + INTERVAL 4 HOUR
WHERE s.stop_id = @new_stop_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = s.vendor_id AND
   ↪  m.user_id = @new_user_id);

UPDATE OperatingHours h SET h.opening_time = '12:00:00', h.closing_time =
 ↪  '20:00:00'
WHERE h.vendor_id = @new_vendor_id AND h.hours_number = 1
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = h.vendor_id AND
   ↪  m.user_id = @new_user_id);

UPDATE OperatingStop s
SET s.actual_start = s.scheduled_start, s.actual_end = s.scheduled_end,
    s.stop_status = 'demo_completed'
WHERE s.stop_id = @new_stop_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = s.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 33
INSERT INTO MenuItem (vendor_id, item_name, description, price, category,
 ↪  availability_status)
SELECT @new_vendor_id, '[DEMO] Practice Dish', 'Disposable learning example',
 ↪  10.00, 'Demo', 'demo_available'
WHERE EXISTS (SELECT 1 FROM Manages WHERE vendor_id = @new_vendor_id AND
 ↪  user_id = @new_user_id);
SET @new_item_id = LAST_INSERT_ID();
UPDATE MenuItem mi SET mi.price = 11.50
WHERE mi.menu_item_id = @new_item_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = mi.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 34
UPDATE Vendor v SET v.description = '[DEMO] Updated description',
 ↪  v.phone_number = '+1-202-555-0199',
    v.website_url = 'https://query-demo.example.invalid'
WHERE v.vendor_id = @new_vendor_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = v.vendor_id AND
   ↪  m.user_id = @new_user_id);
```

```sql
-- Query 35
INSERT INTO Review (author_user_id, vendor_id, rating, review_text,
 ↪  created_at, updated_at)
VALUES (@new_user_id, @new_vendor_id, 4, '[DEMO] Query practice review',
 ↪  UTC_TIMESTAMP(), NULL);
SET @new_review_id = LAST_INSERT_ID();
UPDATE Review SET rating = 5, review_text = '[DEMO] Updated practice review',
 ↪  updated_at = UTC_TIMESTAMP()
WHERE review_id = @new_review_id AND author_user_id = @new_user_id;

-- Query 36
INSERT INTO Photo (uploader_user_id, vendor_id, image_url, caption,
 ↪  uploaded_at)
VALUES (@new_user_id, @new_vendor_id,
        'https://www.themealdb.com/images/media/meals/ustsqw1468250014.jpg',
        '[DEMO] Illustrative pasta photo', UTC_TIMESTAMP());
SET @new_photo_id = LAST_INSERT_ID();
INSERT INTO ReviewPhoto (photo_id, review_id)
SELECT p.photo_id, r.review_id FROM Photo p JOIN Review r ON r.vendor_id =
 ↪  p.vendor_id
WHERE p.photo_id = @new_photo_id AND r.review_id = @new_review_id
  AND p.uploader_user_id = @new_user_id AND r.author_user_id = @new_user_id;
INSERT INTO MenuItemPhoto (photo_id, menu_item_id)
SELECT p.photo_id, mi.menu_item_id FROM Photo p JOIN MenuItem mi ON
 ↪  mi.vendor_id = p.vendor_id
WHERE p.photo_id = @new_photo_id AND mi.menu_item_id = @new_item_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = p.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 37
INSERT INTO Favorites (user_id, vendor_id, saved_at)
VALUES (@new_user_id, @new_vendor_id, UTC_TIMESTAMP())
ON DUPLICATE KEY UPDATE saved_at = Favorites.saved_at;
DELETE FROM Favorites WHERE user_id = @new_user_id AND vendor_id =
 ↪  @new_vendor_id;

-- Query 38
INSERT INTO FoodPermit (vendor_id, permit_number, issuing_authority,
 ↪  permit_type,
                        issued_date, expiration_date, document_url)
SELECT @new_vendor_id, CONCAT('QUERY-DEMO-', @new_vendor_id), 'DEMO AUTHORITY
 ↪  - NOT VALID',
```

```
       'demo_food_service', UTC_DATE(), UTC_DATE() + INTERVAL 1 YEAR, NULL
WHERE EXISTS (SELECT 1 FROM Manages WHERE vendor_id = @new_vendor_id AND
 ↪  user_id = @new_user_id);
SET @new_permit_id = LAST_INSERT_ID();
UPDATE FoodPermit fp SET fp.expiration_date = UTC_DATE() + INTERVAL 2 YEAR
WHERE fp.permit_id = @new_permit_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = fp.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 39
DELETE rp FROM ReviewPhoto rp JOIN Review r ON r.review_id = rp.review_id
WHERE r.review_id = @new_review_id AND r.author_user_id = @new_user_id;
DELETE FROM Review WHERE review_id = @new_review_id AND author_user_id =
 ↪  @new_user_id;

-- Query 40
DELETE mp FROM MenuItemPhoto mp JOIN MenuItem mi ON mi.menu_item_id =
 ↪  mp.menu_item_id
WHERE mi.menu_item_id = @new_item_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = mi.vendor_id AND
   ↪  m.user_id = @new_user_id);
DELETE mi FROM MenuItem mi
WHERE mi.menu_item_id = @new_item_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = mi.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 41
UPDATE Vendor v SET v.listing_status = 'inactive'
WHERE v.vendor_id = @new_vendor_id
  AND EXISTS (SELECT 1 FROM Manages m WHERE m.vendor_id = v.vendor_id AND
   ↪  m.user_id = @new_user_id);

-- Query 42
DELETE FROM ReviewPhoto WHERE photo_id = @new_photo_id;
DELETE FROM MenuItemPhoto WHERE photo_id = @new_photo_id;
DELETE FROM Photo WHERE photo_id = @new_photo_id AND uploader_user_id =
 ↪  @new_user_id;

ROLLBACK;
```

Appendix C: Advanced SQL

These triggers enforce the same-vendor rule for photo attachments.

```
-- Advanced types: existing CHECK constraints and same-vendor TRIGGERS.
-- Multiple triggers count as one advanced type. No table definitions change.
DELIMITER //
USE restaurants//

DROP TRIGGER IF EXISTS check_reviewphoto_insert//
CREATE TRIGGER check_reviewphoto_insert
BEFORE INSERT ON ReviewPhoto
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM Photo p JOIN Review r ON p.vendor_id =
     ↪  r.vendor_id WHERE p.photo_id = NEW.photo_id AND r.review_id =
     ↪  NEW.review_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Photo and Review must
 ↪  belong to the same vendor';
    END IF;
END//

DROP TRIGGER IF EXISTS check_reviewphoto_update//
CREATE TRIGGER check_reviewphoto_update
BEFORE UPDATE ON ReviewPhoto
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM Photo p JOIN Review r ON p.vendor_id =
     ↪  r.vendor_id WHERE p.photo_id = NEW.photo_id AND r.review_id =
     ↪  NEW.review_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Photo and Review must
 ↪  belong to the same vendor';
    END IF;
END//

DROP TRIGGER IF EXISTS check_menuitemphoto_insert//
CREATE TRIGGER check_menuitemphoto_insert
BEFORE INSERT ON MenuItemPhoto
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM Photo p JOIN MenuItem r ON p.vendor_id =
     ↪  r.vendor_id WHERE p.photo_id = NEW.photo_id AND r.menu_item_id =
     ↪  NEW.menu_item_id) THEN
```

```
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Photo and MenuItem must
 ↪  belong to the same vendor';
    END IF;
END//

DROP TRIGGER IF EXISTS check_menuitemphoto_update//
CREATE TRIGGER check_menuitemphoto_update
BEFORE UPDATE ON MenuItemPhoto
FOR EACH ROW
BEGIN
    IF NOT EXISTS (SELECT 1 FROM Photo p JOIN MenuItem r ON p.vendor_id =
     ↪  r.vendor_id WHERE p.photo_id = NEW.photo_id AND r.menu_item_id =
     ↪  NEW.menu_item_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Photo and MenuItem must
 ↪  belong to the same vendor';
    END IF;
END//

DROP TRIGGER IF EXISTS check_photo_vendor_update//
CREATE TRIGGER check_photo_vendor_update
BEFORE UPDATE ON Photo
FOR EACH ROW
BEGIN
    IF EXISTS (SELECT 1 FROM ReviewPhoto rp JOIN Review r ON r.review_id =
     ↪  rp.review_id WHERE rp.photo_id = OLD.photo_id AND r.vendor_id <>
     ↪  NEW.vendor_id) OR EXISTS (SELECT 1 FROM MenuItemPhoto mp JOIN
     ↪  MenuItem m ON m.menu_item_id = mp.menu_item_id WHERE mp.photo_id =
     ↪  OLD.photo_id AND m.vendor_id <> NEW.vendor_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Cannot change photo
 ↪  vendor while attached records use another vendor';
    END IF;
END//

DROP TRIGGER IF EXISTS check_review_vendor_update//
CREATE TRIGGER check_review_vendor_update
BEFORE UPDATE ON Review
FOR EACH ROW
BEGIN
    IF EXISTS (SELECT 1 FROM ReviewPhoto link JOIN Photo p ON p.photo_id =
     ↪  link.photo_id WHERE link.review_id = OLD.review_id AND p.vendor_id <>
     ↪  NEW.vendor_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Cannot change Review
 ↪  vendor while attached photos use another vendor';
```

```
    END IF;
END//

DROP TRIGGER IF EXISTS check_menuitem_vendor_update//
CREATE TRIGGER check_menuitem_vendor_update
BEFORE UPDATE ON MenuItem
FOR EACH ROW
BEGIN
    IF EXISTS (SELECT 1 FROM MenuItemPhoto link JOIN Photo p ON p.photo_id =
     ↪  link.photo_id WHERE link.menu_item_id = OLD.menu_item_id AND
     ↪  p.vendor_id <> NEW.vendor_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Cannot change MenuItem
 ↪  vendor while attached photos use another vendor';
    END IF;
END//

DELIMITER ;
```

Appendix D: Trigger Tests and Recorded Results

These tests check two valid photo links and seven invalid changes. ROLLBACK restores the original data after testing.

```
USE restaurants;
START TRANSACTION;
SET @u=(SELECT user_id FROM User WHERE
 ↪  email='demo.customer.001@example.invalid');
SET @r1=(SELECT MIN(review_id) FROM Review WHERE vendor_id=1);
SET @r2=(SELECT MIN(review_id) FROM Review WHERE vendor_id=2);
SET @m1=(SELECT MIN(menu_item_id) FROM MenuItem WHERE vendor_id=1);
SET @m2=(SELECT MIN(menu_item_id) FROM MenuItem WHERE vendor_id=2);
INSERT INTO Photo (uploader_user_id,vendor_id,image_url,caption,uploaded_at)
VALUES (@u,1,'https://example.invalid/trigger-test.jpg','[DEMO] Rolled-back
 ↪  trigger test',UTC_TIMESTAMP());
SET @p=LAST_INSERT_ID();
INSERT INTO ReviewPhoto(photo_id,review_id) VALUES(@p,@r1);
SELECT 'Valid same-vendor review link',COUNT(*) FROM ReviewPhoto WHERE
 ↪  photo_id=@p;
-- Expected rejection 1: change review attachment to another vendor.
UPDATE ReviewPhoto SET review_id=@r2 WHERE photo_id=@p;
-- Expected rejection 2: change photo vendor while attached to review.
UPDATE Photo SET vendor_id=2 WHERE photo_id=@p;
-- Expected rejection 3: change review vendor while photos are attached.
UPDATE Review SET vendor_id=2 WHERE review_id=@r1;
DELETE FROM ReviewPhoto WHERE photo_id=@p;
-- Expected rejection 4: insert a cross-vendor review attachment.
INSERT INTO ReviewPhoto(photo_id,review_id) VALUES(@p,@r2);
INSERT INTO MenuItemPhoto(photo_id,menu_item_id) VALUES(@p,@m1);
SELECT 'Valid same-vendor menu link',COUNT(*) FROM MenuItemPhoto WHERE
 ↪  photo_id=@p;
-- Expected rejection 5: change menu attachment to another vendor.
UPDATE MenuItemPhoto SET menu_item_id=@m2 WHERE photo_id=@p;
-- Expected rejection 6: change item vendor while a photo is attached.
UPDATE MenuItem SET vendor_id=2 WHERE menu_item_id=@m1;
DELETE FROM MenuItemPhoto WHERE photo_id=@p;
-- Expected rejection 7: insert a cross-vendor menu attachment.
INSERT INTO MenuItemPhoto(photo_id,menu_item_id) VALUES(@p,@m2);
ROLLBACK;
```

Recorded Output

The following output was recorded from the live trigger tests.

```
Valid same-vendor review link   1
Valid same-vendor menu link 1

EXPECTED REJECTIONS:
ERROR 1644 (45000) at line 14: Photo and Review must belong to the same
 ↪  vendor
ERROR 1644 (45000) at line 16: Cannot change photo vendor while attached
 ↪  records use another vendor
ERROR 1644 (45000) at line 18: Cannot change Review vendor while attached
 ↪  photos use another vendor
ERROR 1644 (45000) at line 21: Photo and Review must belong to the same
 ↪  vendor
ERROR 1644 (45000) at line 25: Photo and MenuItem must belong to the same
 ↪  vendor
ERROR 1644 (45000) at line 27: Cannot change MenuItem vendor while attached
 ↪  photos use another vendor
ERROR 1644 (45000) at line 30: Photo and MenuItem must belong to the same
 ↪  vendor
```

A separate invalid-rating test produced:

```
ERROR 3819 (HY000): Check constraint 'chk_rating' is violated.
```

All 16 table counts matched the pre-test counts after rollback.